

# Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

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## ABSTRACT

Stablecoins are designed to trade near one dollar, but stress events repeatedly create visible deviations across venues, quote currencies, and settlement rails. These deviations often appear to be arbitrage opportunities. In practice, however, a quoted price gap is not the same as an executable trade: a trader must clear order-book depth, absorb VWAP slippage, pay fees, and bear settlement risk before any apparent spread becomes profit.

We introduce Stablecoin StressBench, an execution-aware benchmark for evaluating stablecoin dislocation models. The benchmark separates three layers: optical price dislocations, executable arbitrage opportunities, and ex-ante predictable executable opportunities. During the March 2023 USDC/SVB stress window, 34.3% of 1-minute windows exceed a 10 bps max cross-quote basis threshold on price alone, while only 2.88% remain profitable at \$10K notional after VWAP book-walking and trading costs. A hindsight oracle earns positive net bps, confirming that profitable windows exist, but rule-based models, ML classifiers, and a direct expected-net-profit regressor all lose money out of sample.

The result reframes stablecoin arbitrage prediction as an execution-barrier problem rather than a price-signal detection problem. Stablecoin StressBench contributes a historical stress-event catalogue, execution-aware labels, economic evaluation metrics, and a model ladder for future work on crypto market microstructure, optimal execution, and financial benchmark construction.

## KEYWORDS

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

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## 1 INTRODUCTION

Stablecoins are designed to make a simple promise: one token should trade close to one dollar. This promise makes stablecoins central to crypto-market settlement, quoting, collateral, and liquidity transfer. When confidence in that promise weakens, prices fragment across venues, quote currencies, and liquidity pools.

Price fragmentation often appears to create arbitrage. During a stress event, one route may imply that USDC is cheap while another implies that USDT or USD is expensive. A

trader looking only at quoted prices may conclude that buying through one route and selling through another should generate mechanical profit.

That conclusion is incomplete. A quoted price gap is not a trade. To realize arbitrage, a trader must clear available order-book depth, absorb VWAP slippage, pay taker fees, and bear settlement delay. These frictions are small individually but large jointly. As a result, many stablecoin dislocations are optical rather than executable.

This paper introduces Stablecoin StressBench, an execution-aware benchmark for stablecoin dislocations. The benchmark separates three layers: optical price dislocations, executable arbitrage opportunities, and ex-ante predictable executable opportunities. The distinction matters because a model can classify price dislocations accurately while still losing money if the selected windows do not survive execution.

We study this distinction during the March 2023 USDC/SVB stress event and in a broader historical catalogue of stablecoin stress episodes. In the primary Tier-A execution-grade test window, 34.3% of minutes exceed a 10 bps max cross-quote basis threshold, but only 2.88% remain profitable at \$10K notional after VWAP execution and costs. A hindsight oracle confirms that profitable windows exist, yet rule-based models, ML classifiers, and direct expected-profit regressors all fail to capture them out of sample. The result reframes stablecoin arbitrage prediction as an execution-barrier problem: the hard problem is not detecting that a stablecoin price moved, but identifying which dislocations survive execution.

Contributions. We make four contributions.

- (1) Execution-aware stablecoin benchmark. Per-minute labels distinguish quoted price dislocations from executable arbitrage opportunities using order-book depth, VWAP book-walking, fees, notional size, and settlement penalties.
- (2) Historical stress-event framework. We catalogue stablecoin stress events across mechanism classes and classify each event by data availability. Tier-A events support execution-aware labels; Tier-B events support price and liquidity analysis; Tier-C events support historical taxonomy only.
- (3) Price-to-execution and oracle-gap evidence. In the USDC/SVB Tier-A test event, 34.3% of minutes exceed a 10 bps max cross-quote basis threshold, but only 2.88% remain executable at \$10K. A hindsight oracle is profitable on every task; on the executable arbitrage tasks the oracle gap exceeds 260 bps and all deployable models are economically negative.
- (4) Literature-motivated model ladder. We evaluate economic anchors, naive price rules, time-series baselines, regularized linear models, tree ensembles, direct expected-profit

regression, meta-labeling, and regime detectors. This ladder establishes reproducible baseline failures for future models attempting to close the oracle gap.

This paper fits ICAIF as financial benchmark construction for blockchain and cryptocurrency markets, combining market microstructure, trading and execution, validation and calibration of financial models, and economic evaluation of ML systems.

## 2 RELATED WORK

**Stablecoin stress and depegs.** The USDC/SVB episode (March 2023) produced well-documented price fragmentation across venues; we focus on the ex-ante predictability question that event documentation does not address. Kwon et al. [1] analyse Terra/LUNA collapse mechanics; we use May 2022 as our validation split event. Griffin and Shams [2] study Tether price manipulation; our contribution is execution microstructure, not flow manipulation.

**Limits to arbitrage.** Shleifer and Vishny [4] and Gromb and Vayanos [5] establish the general theory. Hautsch et al. [3] model blockchain-specific limits to arbitrage (settlement finality, gas costs, latency); Makarov and Schoar [8] provide empirical evidence of persistent crypto price gaps across venues. We provide an execution-aware empirical measurement of stablecoin arbitrage barriers during a real stress event, using documented order-book depth provenance.

**Market microstructure and optimal execution.** Glosten and Milgrom [7] establish the theoretical basis for adverse selection during stress. Cont and Kukanov [6] develop optimal order-routing across fragmented limit-order books, which motivates our VWAP depth-walk formulation. Adams et al. [9] characterise AMM liquidity dynamics, informing our settlement-penalty modelling.

**Financial ML benchmarks and model validation.** López de Prado [10] introduces meta-labeling to separate primary signal generation from execution filtering directly applicable to our setting where price signals are plentiful but most are not executable. Cintra and Holloway [11] apply BOCPD to Curve StableSwap pool reserves and detect the May 2022 Terra/UST de-peg with a 12-hour lead; we find BOCPD achieves only AUROC 0.23 on CEX cross-quote data, suggesting the pool-reserve signal does not transfer directly. No prior benchmark provides execution-aware stablecoin arbitrage labels with documented route-depth provenance and a systematic price-to-execution gap analysis.

## 3 BENCHMARK DESIGN

### 3.1 Data Sources and Instruments

The gold-layer dataset contains 56,134 1-minute bars across six windows (three calm-control training windows, one validation stress event, one test stress event, and its recovery window) (Table 2). Price and reference features are derived from the Binance Vision public archive (klines, aggTrades) and Coinbase REST candles. Net-profit labels are computed on routes with confirmed order-book depth coverage; the

Table 1: Route-leg depth provenance for the primary cross-quote label.

| Leg   | Instrument | Venue    | SVB test (Mar 2023)      | 2024 windows      |
|-------|------------|----------|--------------------------|-------------------|
| Buy   | BTC-USDC   | Binance  | kline-proxy <sup>†</sup> | real L2 (futures) |
| Cross | USDC-USDT  | Binance  | real L2 (Mar 12–20)      | real L2 (futures) |
| Sell  | BTC-USDT   | Binance  | real L2 (futures)        | real L2 (futures) |
| Ref   | BTC-USD    | Coinbase | candle-proxy             | candle-proxy      |

<sup>†</sup>BTC-USDC perpetual did not exist on Binance before 2024;

kline-proxy used for 2022–2023 windows. 2022 windows use

kline-proxy on all legs (Binance Vision archive does not include futures bookDepth before 2023).

Table 2: Dataset coverage by split.

| Split      | Event                    | Minutes | Pos. rate |
|------------|--------------------------|---------|-----------|
| Train      | Calm control (3 windows) | 28,776  | 0.00%     |
| Validation | Terra/LUNA, May 2022     | 11,526  | 2.30%     |
| Test       | USDC/SVB, Mar 2023       | 15,832  | 2.88%     |
| Total      |                          | 56,134  |           |

Positive rate =  $\text{net\_profit\_bps}_{q10000} > 10 \text{ bps}$ .

cross-quote route uses Binance BTC-USDC and BTC-USDT depth alongside Coinbase BTC-USD as the reference leg.

**Depth provenance and route completeness.** The pipeline tags every depth record with a provenance field: `binance:futures_bookdepth` for real Binance USDM futures bookDepth snapshots, or `binance:klines` / `coinbase:candles` for kline-synthesised approximations. Table 1 summarises per-leg provenance for the primary execution label. The sell leg (BTC-USDT) uses real L2 futures bookDepth for all 2023 and 2024 benchmark windows (Binance Vision public archive). The cross leg (USDC-USDT) uses real L2 futures bookDepth from 2023-03-12 onward, covering 8 of the 10 USDC/SVB test days and all 2024 windows. The buy leg (BTC-USDC) uses kline-synthesised depth for the USDC/SVB test window (the BTC-USDC perpetual did not exist on Binance before 2024) and real L2 futures bookDepth for the 2024 calm-control windows. The net-profit labels reported in this paper are therefore execution-labelled route estimates with documented real/proxy depth provenance, not fully route-complete real-L2 replay on every leg for all windows. A fully route-complete benchmark for 2022 events and the BTCUSDC buy leg in 2023 requires a paid Tardis archive subscription.

### 3.2 Event-Based Split Design

Splits are defined by event identity, not by time alone (Figure 1). The train split contains three calm control windows (no stress event) that bracket the test period; the validation split uses the Terra/LUNA collapse (May 2022), a qualitatively similar but independent stress event; the test split uses the USDC/SVB de-peg (March 10–15, 2023).

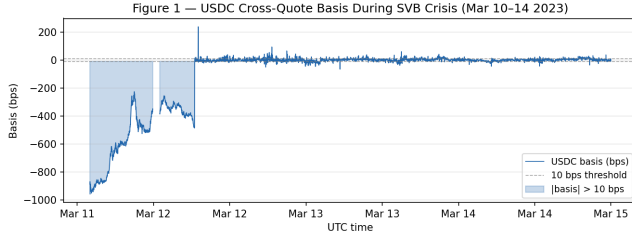


Figure 1: Cross-quote basis (bps) during the SVB stress window (March 2023), exceeding 100 bps for sustained periods. With real USDC/USDT futures bookDepth, the USDC-specific basis exceeds 10 bps in 12.5% of test minutes (capturing the actual SVB-period USDC de-peg); the USDT route exceeds 10 bps in 27.4% of minutes; the max-abs basis (maximum of both routes) exceeds 10 bps in 34.3%.

Table 3: Data tier and permitted claims.

| Tier     | Data                                                                             |
|----------|----------------------------------------------------------------------------------|
| A (N=2)  | Frozen execution labels with documented route-depth provenance at execution time |
| B (N=11) | OHLCV / pool data                                                                |
| C (N=5)  | Partial / DEX-only                                                               |

This design has two guarantees: (i) no model trained on the train split has seen stress dynamics; (ii) threshold calibration on validation uses an independent stress event, not a held-out portion of the same episode. No lookahead: labels are constructed by `join_asof` at  $t + h$  and formally verified. The Terra/LUNA validation split is not used to claim a second execution-grade test result; its role is calibration, exposing the thresholding procedure to independent stress dynamics before final evaluation on USDC/SVB.

**3.3 Historical Stress Taxonomy and Data Tiering**  
Stablecoin stress is not a single phenomenon. We catalogue 18 stress events from 2020–2023 spanning seven distinct failure mechanisms: algorithmic/reflexive (FEI, IRON/TITAN, MIM, UST, USDD), fiat-reserve bank shock (USDC/SVB), regulatory/issuer winddown (BUSD, Binance conversion), exchange credit/liquidity (Celsius/3AC, HUSD, FTX [12]), DeFi pool imbalance (Curve/UST, USDT/Curve, USDC/DAI secondary), collateral liquidation (DAI Black Thursday), and RWA/niche stablecoin (Acala aUSD, USDR).

Events are classified by data availability tier, which governs what claims this paper may make (Table 3). Execution-grade labels are available for the USDC/SVB stress window and its immediate post-stress recovery window (both Tier A). The price-to-execution gap, oracle net bps, and model evaluation in this paper are anchored to the USDC/SVB

Table 4: Stress-event roles in Stablecoin StressBench.

| Event           | Mechanism               | Tier       | Role                      |
|-----------------|-------------------------|------------|---------------------------|
| Terra/UST 2022  | Algorithmic/reflexive   | Validation | Threshold calibration     |
| USDC/SVB 2023   | Fiat-reserve bank shock | A          | Main execution test       |
| USDC recovery   | Fiat-reserve recovery   | A          | Comparator                |
| USDT/Curve 2023 | DeFi pool imbalance     | B          | Price/liquidity context   |
| FTX 2022        | Exchange credit/liq.    | B          | Market stress context     |
| BUSD 2023       | Regulatory winddown     | B          | Issuer/regulatory context |
| IRON/TITAN 2021 | Algorithmic/reflexive   | C          | Taxonomy context          |

stress test; the recovery window serves as a Tier-A comparator, not an independent stress mechanism. Price-grade claims for Tier-B events are labelled as estimates (est.) and are excluded from any arbitrage-profitability argument.

L2 data availability is not equivalent to execution-grade coverage. A stablecoin arbitrage label requires route-complete depth: synchronized buy- and sell-leg order books, sufficient depth at the tested cap/tilt, and non-synthetic provenance for the relevant venue (subject to relevant venue caveats). For Tier-B and Tier-C events (FTX, Celsius/3AC, USDT/Curve, BUSD, and others) no order-book depth data is available in the repository. Those events support price and taxonomy claims only (see docs/execution\_route\_coverage.md).

### 3.4 Stress-Event Roles in the Benchmark

Table 4 clarifies how the historical catalogue is used. The catalogue establishes that stablecoin stress recurs across mechanisms, but only Tier-A events support execution-aware P&L claims. USDC/SVB is the primary execution-grade test; USDC recovery is a Tier-A comparator; and Tier-B / Tier-C events provide external validity and mechanism taxonomy rather than arbitrage-profitability evidence.

Terra/UST serves as the validation split (Tier B, Algorithmic/reflexive mechanism): it is used for threshold calibration and exposes the thresholding procedure to independent stress dynamics before final evaluation. It is not used to claim a second execution-grade result the validation split has executable-arb labels (2.30% net-profitable at \$10K, 10bps threshold) but lacks the full Tier-A L2 depth reconstruction required for execution P&L inference.

### 3.5 Feature Sets

We define four nested feature sets from narrowest to broadest:

|                                |                                                   |
|--------------------------------|---------------------------------------------------|
| <code>price_only</code>        | 5 cross-quote basis columns                       |
| <code>price_plus_book</code>   | +7 microstructure cols (spread, depth, imbalance) |
| <code>price_book_frag</code>   | +3 cross-venue fragmentation cols                 |
| <code>price_book_settle</code> | +7 on-chain settlement proxies                    |

The nesting structure isolates the marginal value of each information layer.

Table 5: Three layers of stablecoin dislocation measured by Stablecoin StressBench (USDC/SVB test split, \$10K notional).

| Layer                     | What it measures                           | SVB net         |
|---------------------------|--------------------------------------------|-----------------|
| 1. Optical dislocation    | Quoted basis >10 bps on price data         | 34.3%           |
| 2. Executable opportunity | Net profit >10 bps after VWAP, fees, costs | 2.88%           |
| 3. Predictable ex ante    | Model identifies window before execution   | No active model |

Layer 3 frequency: every active non-oracle strategy that takes trades produces negative net bps; abstention (no-trade) is the zero-P&L floor.

## 4 EXECUTION-AWARE LABELS

The benchmark measures three layers of stablecoin dislocation (Table 5). Understanding these layers clarifies why standard price-signal metrics are insufficient: a model can do well on Layer 1 while failing economically at Layer 2. The gap between Layer 1 and Layer 2 is the price-to-execution gap; the gap between Layer 2 and Layer 3 is the oracle gap. Both are central measurements of this benchmark.

### 4.1 Cross-Quote Basis

The cross-quote USDC basis at time  $t$  is:

$$b_{\text{USDC}}(t) = 10,000 \times \frac{P_{\text{BTC-via-USDC}}(t) - P_{\text{BTC-direct}}(t)}{P_{\text{BTC-direct}}(t)} \quad [\text{bps}]$$

where  $P_{\text{BTC-via-USDC}}$  is the effective BTC price obtained by converting USD  $\rightarrow$  USDC  $\rightarrow$  BTC relative to the direct BTC-USD market. We also compute the max-absolute basis  $b_{\text{max}}(t) = \max(|b_{\text{USDC}}|, |b_{\text{USD}}|)$  as a broader stress detector.

Basis labels for horizon  $h \in \{1, 5, 15, 60\}m$  and threshold  $\tau$ :

$$y_{\text{basis}}(t) = 1[|b_{\text{USDC}}(t+h)| > \tau]$$

Primary classification task: `basis_usdc_1m_gt10bps`.

### 4.2 VWAP Execution Walk

For each minute window and notional  $q \in \{\$10K, \$50K, \$100K, \$500K\}$  we simulate a round-trip execution:

- (1) Walk the available execution-grade bid and ask ladders for routes with confirmed real-L2 provenance to fill  $q$  notional, recording the VWAP price at each level consumed.
- (2) Compute the gross spread:  $\text{gross}(q) = \text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}}$ .
- (3) Deduct taker fees on both legs and a settlement delay penalty.

### 4.3 Net Executable Arbitrage Labels

$$\text{net}(q, t) = \text{gross}(q, t) - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}} \quad (1)$$

$$y_{\text{arb}}(q, h, t) = 1 \left[ \max_{s \in [t, t+h]} \text{net}(q, s) > 0 \right] \quad (2)$$

where  $f_{\text{buy}}, f_{\text{sell}}$  are venue taker fees and  $\delta_{\text{settle}}$  is a settlement penalty proxy. Primary economic task: `executable_arb_q10000_5m`. The forward-max in Eq. (2) asks: does a profitable window exist within the next  $h$  minutes? matching the practical

constraint that a trader can execute at any point within the horizon.

## 4.4 Oracle Upper Bound

The oracle trades every minute where  $\text{net}(q, t) > 0$  in hindsight. It uses information unavailable at prediction time and sets the theoretical performance ceiling. The oracle is not deployable; it exists to confirm that profitable windows exist and to calibrate the oracle gap.

## 5 MODELS AND EVALUATION

### 5.1 Model Ladder

The model stack is a ladder of increasingly realistic baselines, progressing from economic anchors through naive price rules, time-series persistence, interpretable linear models, nonlinear tabular models, direct expected-profit regression, and finance-specific filters. The purpose is not to exhaust all possible architectures; it is to establish whether progressively stronger and more economically aligned models can close the oracle gap.

Economic anchors. `NoTrade` (the economic floor: 0 net bps) and `Oracle` (hindsight ceiling).

Rule baselines. `PriceBasis10bps` (trade when  $|b_{\text{USDC}}| > 10$  bps), `PriceBasis25bps`, and `GrossArbThreshold` (gross spread  $> 0$ ). These test whether threshold rules on price data are sufficient; motivated by [3], execution frictions are expected to make raw price rules uneconomic.

Statistical baselines. `LastValue` (AR0), `RollingMean`, AR1 testing whether basis autocorrelation alone identifies executable stress. Each baseline uses an explicitly named lag column rather than assuming column ordering.

ML classifiers. `Logistic` (L2), `Lasso` (L1), `Random Forest`, `XGBoost`, and `LightGBM` each trained on the four nested feature sets. Tree ensembles are included for their strong tabular performance in financial time-series settings [8].

Expected-net-profit regressor. `ExpectedNetProfitRegressor` directly predicts  $\text{net}(q, t)$  and trades when  $\hat{y} > \theta^*$ . If this also fails, the oracle gap is structural, not a label-mismatch artefact.

Meta-labeling filter. Motivated by [10]: the primary signal is  $|b_{\text{USDC}}| > 10$  bps; a secondary `LightGBM` meta-model learns  $\hat{y}_{\text{meta}} = \Pr[\text{net}(q, t) > 0 \mid \text{primary fires}]$  and filters predicted false positives.

Regime detectors. EWMA z-score, CUSUM, and BOCPD [11] identify stress regimes from the basis time series as two-stage precursors: detect stress first, then apply execution filters only during detected regimes.

Table 6: Price-to-execution gap (test split, \$10K notional, 1-minute same-window horizon).

| Thresh. | Price (max) | Price (USDC) | Exec. (\$10K) | Ratio |
|---------|-------------|--------------|---------------|-------|
| 0 bps   | 95.86%      | 82.96%       | 3.34%         | 29×   |
| 5 bps   | 61.99%      | 25.32%       | 3.03%         | 20×   |
| 10 bps  | 34.33%      | 12.45%       | 2.88%         | 12×   |
| 25 bps  | 9.54%       | 6.48%        | 2.62%         | 3.6×  |

## 5.2 Threshold Calibration

For probabilistic classifiers, the decision threshold is calibrated on the validation split:

$$\theta^* = \arg \max_{\theta \in [0.05, 0.95]} \sum_{i: \hat{p}_i > \theta} \text{net}(q, t_i) \quad \text{s.t.} \quad |\{i : \hat{p}_i > \theta\}| \geq 25$$

The  $\geq 25$  trade minimum prevents degenerate abstention strategies from winning on small-sample artefacts.

## 5.3 Evaluation Protocol

Primary (economic): `net_bps_captured` (mean net bps per trade), `hit_rate_above_cost`, `final_pnl_usd`, `oracle_capture_pct` (= net bps / oracle net bps).

Secondary (statistical): AUROC, AUPRC, Brier score,  $n_{\text{trades}}$ .

The NoTrade baseline (= 0 net bps) is the economic floor: a model that loses money is worse than abstaining entirely, regardless of its AUROC.

# 6 RESULTS

## 6.1 Price Gaps Are Common, Execution Is Rare

Table 6 and Figure 2 show the central finding. At a 10 bps threshold, 34.3% of test-split minutes show a max cross-quote basis exceeding the threshold on price alone; only 2.88% yield net profit above 10 bps after execution costs at \$10K notional a 12× price-to-execution ratio. With real USDCUSDT futures bookDepth, the USDC-specific basis fires in 12.5% of minutes (correctly capturing the SVB-period USDC discount); the USDT route contributes the majority of the max-abs signal (USDT >10 bps in 27.4% of minutes), with USDC dominating the max-abs signal in 34.3% of those minutes. The gap persists at every threshold level, widening sharply below 10 bps as price signals fire constantly but order-book depth is thin.

Tracing the pipeline stage by stage: of 15,832 total test minutes, 5,435 (34.3%) show a max-basis price dislocation above 10 bps; 456 of those (2.88%) yield net profit above 10 bps at \$10K/1m; the oracle trades 315 of those minutes profitably. Naive price rules trade far more often and lose money on nearly all trades; calibrated ML models trade less aggressively or abstain entirely, but none achieves positive economic performance.

## 6.2 Profitable Windows Exist, But Only in Hindsight

Table 7 gives the oracle gap for each primary task. The oracle earns +162.2bps on the `basis_usdc_1m_gt10bps` task (315

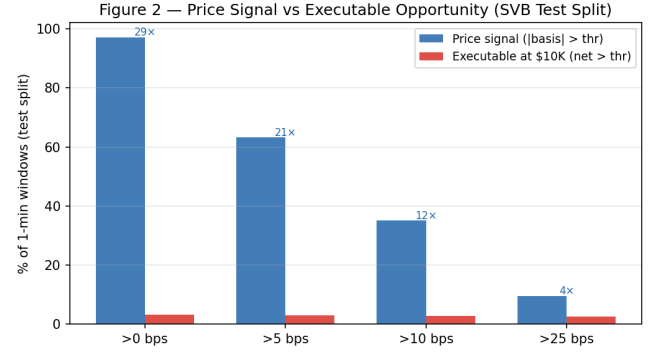


Figure 2: Price-to-execution gap by threshold. Price-signal frequency (dark) dwarfs executable frequency (light) at every level. The gap is largest at low thresholds, where price signals fire constantly but depth is thin.

Table 7: Oracle gap (test split). All non-oracle models are economically negative.

| Task                                  | Oracle     | Best model | Gap     |
|---------------------------------------|------------|------------|---------|
| <code>basis_usdc_1m_gt10bps</code>    | +162.2 bps | +17.2 bps* | 145 bps |
| <code>executable_arb_q10000_5m</code> | +225.0 bps | -41.1 bps  | 266 bps |
| <code>executable_arb_q50000_5m</code> | +147.1 bps | -113.0 bps | 260 bps |

\*lgbm@all features (106 trades); marginal 145 bps below oracle.

Best model: `price_threshold_25bps@price_only` (arb tasks).

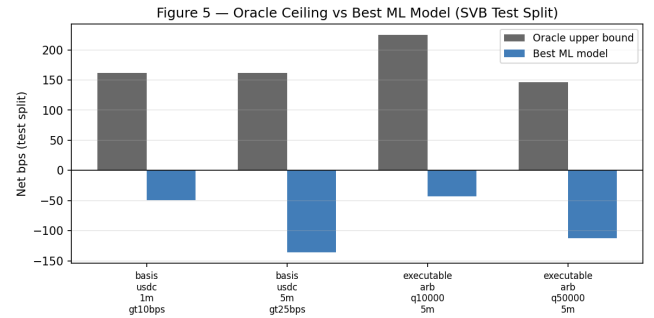


Figure 3: Oracle gap: grouped bars showing oracle net bps (gold) vs. best ML model (navy) for each task. The gap exceeds 200 bps across all tasks.

trades). The best non-oracle model, LightGBM with full features, achieves +17.2bps on this task a 145 bps gap from the oracle. On the executable arbitrage task, the oracle earns +225.0bps; the best rule (`price_threshold_25bps`) achieves -41.1 bps, a 266 bps gap. On the executable arbitrage tasks, every active non-oracle strategy produces negative net bps; the oracle gap is structurally large regardless of cost regime. The LightGBM result on the basis classification task is marginal (17.2bps, 106 trades) and does not close the oracle gap.

Table 8: Model ladder summary (task: **basis\_usdc\_1m\_gt10bps**, test split). <sup>†</sup>Theoretical ceiling.

| Model               | Features   | Net bps | Trades | AUROC |
|---------------------|------------|---------|--------|-------|
| Oracle <sup>†</sup> |            | +162.2  | 315    |       |
| lgbm                | all        | +17.2   | 106    | 0.717 |
| logistic            | all        | −75.8   | 648    | 0.143 |
| gross_arb           | price_only | −136.4  | 607    | 0.531 |
| price_thr 10        | price_only | −270.0  | 1969   | 0.833 |
| price_thr 25        | price_only | −347.3  | 1025   | 0.746 |
| no_trade            |            | 0.0     | 0      |       |
| ExpNetProfit        | price_only | −73.8   | 537    |       |

Table 9: Price-to-execution ratio across cost regimes (\$10K, 10 bps, 5-min horizon, test split).

| Settlement | Fee regime          | Exec. % | Ratio |
|------------|---------------------|---------|-------|
| 0 bps      | low (institutional) | 6.11%   | 2.07× |
| 0 bps      | base                | 5.64%   | 2.24× |
| 0 bps      | high                | 5.46%   | 2.32× |
| 5 bps      | base                | 5.16%   | 2.45× |
| 10 bps     | base                | 4.88%   | 2.59× |
| 10 bps     | high (worst case)   | 4.81%   | 2.63× |

### 6.3 The Model Ladder Fails Economically

Table 8 summarises the model ladder. Price-threshold rules overtrade severely (607–1969 trades, −136 to −347 bps): the rules fire on price dislocations with no executable depth. Validation-thresholded logistic takes 648 trades at −75.8 bps. LightGBM with full features achieves +17.2 bps (106 trades, AUROC 0.72), the only non-oracle result above zero. However, this represents only 106 high-confidence predictions against an oracle that identifies 315 profitable windows earning +162.2 bps each; the 145 bps oracle gap remains. On the executable arbitrage tasks (which target net profit directly), no model crosses zero – the oracle gap there exceeds 260 bps.

The ExpectedNetProfitRegressor (target:  $\text{net}(q, t)$ ) achieves −73.8 bps negative despite directly regressing on the economic criterion used for evaluation. This is the strongest evidence that the oracle gap is structural: even when the training objective directly matches the economic evaluation criterion, the model cannot identify profitable windows out of sample.

### 6.4 Robustness and Diagnostics

Cost-regime robustness. The price-to-execution gap persists across all tested cost assumptions (Table 9). At base fees, 5.64% of windows are executable at \$10K/5m/10 bps; high fees reduce this to 5.46%. Adding a 10 bps settlement penalty gives 4.88%. At \$50K notional the executable rate drops to 1.62%; at \$500K, virtually no windows are executable (0.03%). The conclusion is qualitatively identical across all regimes.

Table 10: Regime detection (test split, stress =  $|b_{\text{USDC}}| > 10 \text{ bps}$ ).

| Detector                 | Acc.  | Recall | FAR   | AUROC |
|--------------------------|-------|--------|-------|-------|
| EWMA (span=20, thr=2.5)  | 0.869 | 0.010  | 0.007 | 0.624 |
| CUSUM (k=0.5, h=5)       | 0.270 | 1.000  | 0.835 | 0.582 |
| BOCPD (hz=0.01, thr=0.5) | 0.860 | 0.022  | 0.019 | 0.229 |

FAR = false alarm rate.

Meta-labeling null result. The MetaLabelingFilter is trained on windows where the primary price signal fires ( $|b_{\text{USDC}}| > 10 \text{ bps}$ ). In the train split only 53 such windows exist, and zero are net-profitable – consistent with the zero positive rate in calm-control training data (Table 2). The meta-model learns no positive class signal and takes 0 trades on the test split. This is an informative null: meta-labeling is architecturally sound but requires a training regime with both positive and negative meta-labels. It motivates using the Terra/LUNA validation split for meta-model training in future work.

Regime detectors. Table 10 shows the recall-precision trade-off. CUSUM achieves near-perfect recall (0.9995) at 83.5% false alarm rate; EWMA achieves 86.9% accuracy at 0.65% false alarm rate with very low recall (0.95%); BOCPD achieves only AUROC 0.229 on the CEX cross-quote series. No detector alone solves the execution identification problem; their value is as a pre-filter that reduces the search space.

False-positive diagnosis. Applying PriceBasis10bps to the test split produces 2,002 trades. False positives 581 windows where basis > 10 bps but order-book depth is insufficient – account for 29% of trades at a mean net loss of −38.7 bps. This directly illustrates the execution-barrier problem: the rule fires on price dislocations where the book is too thin to fill the arbitrage leg.

### 6.5 Stress Recurs Across Mechanisms, But Execution Labels Are Scarce

The benchmark is evaluated on a single Tier-A stress event, which raises the question of whether the price-to-execution gap and oracle gap are specific to the USDC/SVB episode. The historical catalogue (Table 4) provides indirect evidence that execution barriers are a structural property of stablecoin stress rather than an artefact of one event.

Across seven mechanism classes, stress events share common features: a price dislocation arises rapidly, order-book depth deteriorates at the moment when arbitrage would be most profitable, and settlement risk rises simultaneously. This pattern holds regardless of whether the stress originates from a bank-reserve shock (USDC/SVB), an algorithmic-reflexive collapse (Terra/UST), an exchange credit event (FTX), a DeFi pool imbalance (USDT/Curve), or a regulatory wind-down (BUSD).

The constraint is not whether stress occurs it does, repeatedly but whether execution-grade order-book data exists to measure the gap precisely. Only Tier-A events provide the L2 depth required for VWAP label construction. The Terra/LUNA validation split ( $N=11,526$  minutes, 13.46% price-basis positive, 2.30% executable at \$10K) demonstrates similar dynamics under an independent mechanism, but without the full Tier-A execution reconstruction needed for P&L claims. Expanding Tier-A coverage to additional mechanism classes is the primary path to testing whether the oracle gap generalises across stress regimes.

## 7 LIMITATIONS

Tier-A execution-grade history is scarce. The test split covers one execution-grade stress event (SVB March 2023). Future work should extend Tier-A coverage to additional mechanism classes; the 18-event catalogue documents the data requirements. Models trained on fiat-reserve bank shocks may not generalise to algorithmic-reflexive or exchange-credit failure modes.

Historical catalogue is broader than benchmark. Tier-B and Tier-C events support price and taxonomy claims only. The benchmark deliberately separates what can be claimed from Tier-A data from what can only be described at Tier-B or Tier-C.

Settlement and latency are proxied. CEX-to-CEX settlement latency is approximated via a fixed penalty parameter; true on-chain settlement delay and block confirmation times are not modelled. A more detailed treatment would require per-block latency data beyond the scope of this benchmark. Venue-depth coverage. Binance USDM futures bookDepth for BTC-USDT is included in the public release for 2023 and 2024 benchmark windows, providing real L2 depth for the sell leg of the cross-quote route. The BTC-USDC buy leg and all 2022 windows use kline-synthesised depth (the 2022 Binance Vision archive does not include futures bookDepth). Full historical L2 for the Coinbase BTC-USD leg requires a Tardis subscription and is not committed to the public repository.

Model scope. The ladder covers tabular ML and statistics-based approaches. Deep sequence models, graph networks, and RL-based order-execution policies are outside scope and represent natural extensions. The oracle gap defines the challenge they must close.

## 8 CONCLUSION

Stablecoins promise price stability, but stress events create visible dislocations across venues. Those dislocations look like arbitrage on a chart. The central result of this paper is that they mostly are not: 34.3% of minutes in the USDC/SVB stress window show a cross-quote basis above 10 bps, but only 2.88% remain net-profitable after VWAP book-walking and costs at \$10K notional a 12 $\times$  price-to-execution gap.

Profitable windows do exist: a hindsight oracle earns +162.2 bps per trade on the USDC-basis task, confirming that execution-grade opportunities are real, not artefacts of label construction. On the executable arbitrage tasks every active non-oracle strategy price rules, ML classifiers, tree ensembles, and a direct expected-net-profit regressor produces negative net bps out of sample; abstention remains the zero-P&L floor. The oracle gap exceeds 260 bps on the executable arbitrage tasks and persists across fee regimes, settlement penalties, and notional sizes.

Stablecoin StressBench shows that the main challenge is not observing price dislocations, but predicting which dislocations survive execution. The benchmark defines an oracle gap between hindsight-executable opportunities and ex-ante deployable models. The historical catalogue shows why this benchmark matters beyond one event: stablecoin stress recurs across mechanisms, while only some episodes provide the depth required for execution-aware evaluation. Closing the oracle gap through better microstructure features, improved calibration, or deeper sequence and network models is the central challenge for future work on crypto market microstructure, financial ML validation, and execution-aware trading systems.

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